INCH-POUND

MS25325R
25 August 2020
SUPERSEDING
MS25325P
27 November 2003

RELAYS, ELECTROMAGNETIC, 5 AMPERES, 4 PDT, TYPE I, SOCKET MOUNTED, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 5 JUNE 1987.
NO SUPERSEDING SPECIFICATION
FOR NEW DESIGN USE MIL-PRF-83536/5 OR /6

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

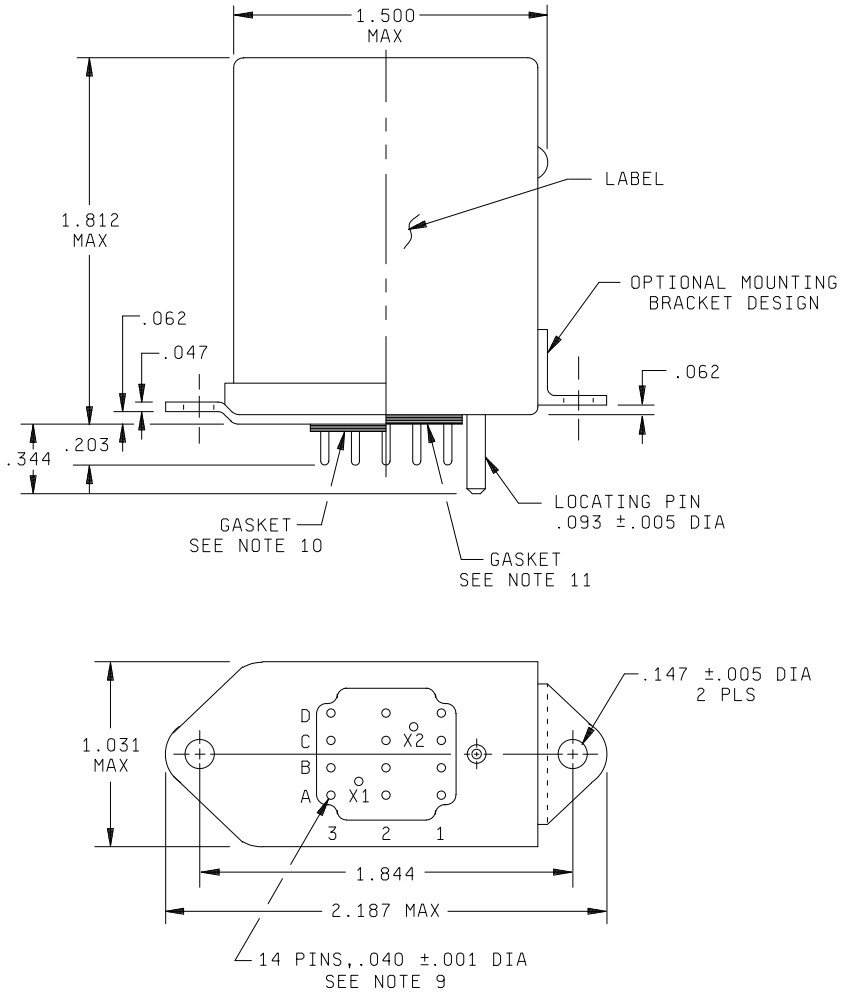
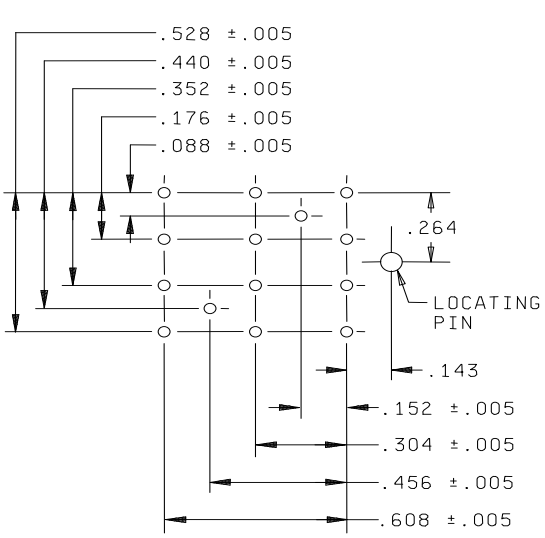
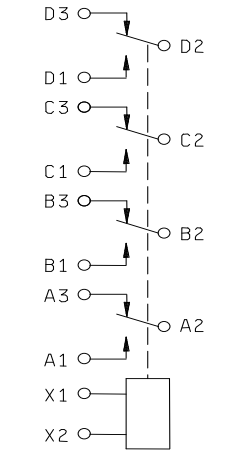


FIGURE 1. Dimensions and configurations.

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CIRCUIT DIAGRAM
SEE NOTES 4 AND 5

Inches	mm	Inches	mm
.001	0.03	.203	5.16
.005	0.13	.264	6.71
.008	0.20	.304	7.72
.010	0.25	.344	8.74
.040	1.02	.352	8.94
.047	1.19	.440	11.18
.062	1.57	.456	11.58
.075	1.91	.528	13.41
.088	2.24	.608	15.44
.093	2.36	1.031	26.19
.143	3.63	1.500	38.10
.147	3.73	1.812	46.02
.152	3.86	1.844	46.84
.176	4.47	2.187	55.55

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers need not appear on relay headers provided there is affixed positively identifies each terminal location specified herein.
5. The use of diodes on ac relays is optional. Actual application must be shown on label.
6. Pins to be perpendicular to header surface within one degree.
7. In the event of conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
8. Referenced Government documents of the issue listed in Assist Online (https://assist.dla.mil/) or Assist Quick Search (https://quicksearch.dla.mil) specified in the solicitation form a part of this specification to the extent specified herein.
9. Socket pin terminals shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold-plated contacts. Terminals shall be gold plated. One system for gold plating that may be used is ASTM-B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microinches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.

FIGURE 1. Dimensions and configurations - Continued.

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10. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have thickness .075, recess .047 in bracket ±.005. Gasket material according to SAE-AMS3332 has been considered acceptable.
11. Gasket shall provide a reliable seal between the relay and mating socket that will meet the environmental, operational, and interface requirements of the relay with the mating socket. The gasket shall have shore hardness 20 ±5, thickness .047 ±.008. Gasket material according to SAE-AMS3332 has been considered acceptable.

FIGURE 1. Dimensions and configurations - Continued.

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TABLE I. Dash numbers and characteristics.

Dash number MS25325-	Type	Coil	Terminal type	Max weight in pounds
D1	I	dc	Plug in	0.29
A1	I	ac	Plug in	0.30
A2	I	ac	Plug in	0.30

TABLE II. Operating characteristics.

PIN MS 25325-	Coil data											Time - (milliseconds maximum)					
	Coil	Nominal			Max		Max pick-up voltage			Hold voltage 2/	Drop out voltage 2/	Operate 3/	Release 4/	Contact Bounce			
														Main		Aux	
		Volts 1/	Freq Hz	Res Ω	Volts	Amp	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
D1	X1,X2	28	dc	N/A	29	0.13	18	19.8	22.5	7.0	1.5	20	20	2	2	---	---
A1	X1,X2	115	400	N/A	122	0.06	90	95	100	35	5.0	25	50	2	2	---	---
A2	X1,X2	115	50/ 60 400	N/A	122	0.07	90	95	100	35	5	25	50	2	2	---	---

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	5	5			5	4							
Inductive	100													
Inductive	20	3	3			3	2							
Motor	100	1.5	1.5			1.5	1							
Lamp	100	0.8	0.8			0.8	0.6							
Transfer load														2/
Mechanical life reduced current	400	1.25	1.25			1.25	1							
Mixed loads	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Environmental characteristics.

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 μs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1,500 Hz
Acceleration	15 G

Electrical characteristics.

Minimum insulation resistance, initial	100 megohms.
After life or environmental tests	50 megohms.

Dielectric strength (sea level).

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts	N/A	N/A
All other points	1,000 V rms	1,000 V rms

Dielectric strength (altitude):

	80,000 ft
Coil to case	500 V rms
Aux contacts	
All other points	500 V rms

Max contact drop initial:	0.150 volt.
After life test:	0.175 volt.
Overload current (NO):	20 amperes.
Rupture current	25 amperes.
Duty rating:	Continuous.
RFI specification:	MIL-STD-461.

(Applicable to coil circuits of ac operated relays).

Conformance inspection.

Performance of groups B and C tests are not applicable.

Group A acceptance reports shall be submitted to the preparing activity on a yearly basis in order to retain qualification for this military standard sheet.

Qualification by similarity: See MIL-PRF-6106.

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The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

ASTM-B488 MIL-STD-461 SAE-AMS3332

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2020-042)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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